

IMBANG: Balancing DDoS Mitigation and Service Outages with Safety-Spanning Group-Relative Policy Optimisation

Author One, Author Two, and Author Three

Abstract—Sentinel combines PPO, a co-evolving attacker, and symbolic shielding for DDoS mitigation, yet severe outages persist under floods. We replace this loop with critic-free Group Relative Policy Optimization (GRPO) and show that group centring removes the marginal influence of an outage penalty whenever all rollouts share the same outage count. Naive GRPO reduces leakage while increasing outages. IMBANG adds one guaranteed no-less-safe rollout per group with an outage budget. Over ten seeds, IMBANG achieves higher composite scores than Sentinel in all eight scenarios, reduces severe outages by 57%, and meets the outage budget in nine seeds.

Index Terms—DDoS mitigation, 6G security, safe reinforcement learning, group-relative policy optimisation.

I. INTRODUCTION

ZERO-TOUCH 6G management expects edge controllers to absorb volumetric DDoS floods in closed loop [1], [2]. Such a controller must discard hostile traffic without starving legitimate users: a filter that is too aggressive causes the outage it was meant to prevent [3]. Deep reinforcement learning has been applied to throttling, SDN flow control and per-host drop policies [4]–[6]. Sentinel [7] is the most complete recent design. It trains its agent with Proximal Policy Optimization (PPO) [8] against a co-evolving attacker and inserts a symbolic shield [9] between the agent and the network. Its authors report that training deteriorates late in every seed and that severe outages persist under heavy floods.

Group Relative Policy Optimization (GRPO) [10] looks like a natural successor. It compares several rollouts started from one state, so neither a value network nor a learned opponent is needed. In a simulator these rollouts can share one realisation of the traffic process [11], so that they differ only in the defender’s actions. We find, however, that GRPO defenders leak less attack traffic than Sentinel but cause *more* severe outages. The reason is structural. A cost that every member of a group incurs is removed by the group mean, mirroring the gradient starvation of binary-reward GRPO [12], [13]. Separate normalisation of the cost channel, as in Constrained GRPO [14], does not help when that channel has no variance within a group. Injecting reference samples into groups stabilises language-model training [15], [16], but no safe direction is available in that setting. Table I summarises how these approaches treat the outage constraint.

This letter makes three contributions. First, it formalises this *constraint-cancellation* effect and confirms it on 80 trained

TABLE I
TREATMENT OF THE OUTAGE CONSTRAINT BY RELATED APPROACHES

Approach	Critic-free	No learned attacker	Outage budget	Cost signal when all members violate
Sentinel [7]	–	–	–	n/a (critic)
PPO-Lagrangian / CPO [17]	–	✓	✓	n/a (critic)
GRPO [10] with penalty	✓	✓	✓	–
Constrained GRPO [14]	✓	✓	✓	–
Sign advantage [12]	✓	✓	–	fixed reference
IMBANG (ours)	✓	✓	✓	✓ (safe member)

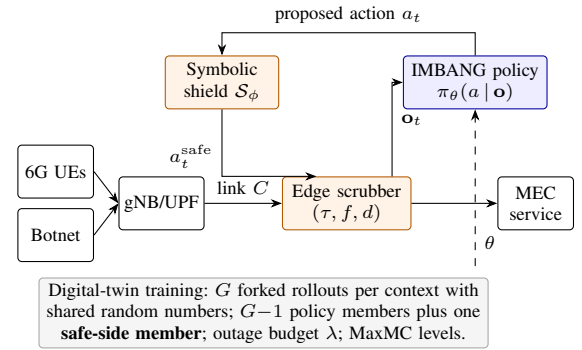


Fig. 1. System model. At the 6G edge, a learned filtering policy configures a scrubber placed in front of the MEC service, and every proposal passes through a symbolic shield. The policy is trained in a digital twin using safety-spanning rollout groups.

checkpoints. Second, it proposes IMBANG (Interpretable, Monotone-safe, Budgeted, Adversary-free, Neuro-symbolic GRPO; also Malay for *balance*), whose *safe-side member* keeps every group straddling the outage boundary, with a monotonicity guarantee. Third, it compares IMBANG with Sentinel’s released checkpoints, a compute-matched PPO baseline, Constrained GRPO and the simplest alternative remedies over ten seeds.

II. SYSTEM MODEL

We use the setting of [7], illustrated in Fig. 1. An edge link of capacity C carries benign traffic $L_t \sim \text{Poisson}(\mu)$ and attack traffic $A_t = 1.5 C i_t$. The attack belongs to a family in $\{\text{SYN}, \text{UDP}, \text{HTTP}, \text{ICMP}, \text{mixed}\}$, with intensity $i_t \in [0, 1]$ and a mutation level m_t that weakens filter matching; flash crowds may inflate L_t up to fourfold. ICMP is withheld from training. In each control window the defender observes twelve aggregate indicators $\mathbf{o}_t \in [0, 1]^{12}$, measured on served traffic. It then sets a load threshold τ_t , a filtering focus $f_t \in$

$\{\text{SrcIP}, \text{Proto}, \text{Size}\}$ and a drop probability d_t . With match effectiveness $\eta = \eta_{kf}(1 - m_t/2)$, the share of benign traffic lost to filtering is

$$\text{CD}_t = 0.4 d_t(1 - \eta) \alpha(\tau_t) + 0.2 [L_t - C\tau_t]^+ / L_t, \quad (1)$$

where $\alpha(\tau)$ is the fraction of offered load above the threshold. Service quality is $\text{SQ}_t = \rho_t(1 - \text{CD}_t)$, with congestion factor $\rho_t = \min\{1, C/(L_t + A_t)\}$. A window with $\text{SQ}_t < 0.5$ is a *severe outage*. The operator scores each window with Sentinel's benchmark, $b_t = \text{SQ}_t - \text{leak}_t - 0.5 \mathbf{1}[\text{SQ}_t < 0.5] - 0.25 \mathbf{1}[\text{SQ}_t < 0.75]$, where leak_t is the share of attack traffic delivered. A symbolic shield $\mathcal{S}_\phi$ edits unsafe actions through per-coordinate clamps. The defender solves

$$\max_{\theta} \mathbb{E}_{\pi_\theta} \left[\sum_t b_t \right] \quad \text{s.t.} \quad \sigma(\pi_\theta) \leq \kappa = 0.05, \quad (2)$$

where σ is the severe-outage frequency on the operator's validation scenarios. The budget κ was fixed before the ten-seed experiments.

III. CONSTRAINT CANCELLATION

GRPO copies each training context into G members that replay the same traffic. Member g collects objective $O_g = \sum_h b_h$ and outage count K_g over H windows. With a Lagrange multiplier λ , the return is $R_g = O_g - \lambda K_g$ and the advantage is $\hat{A}_g = (R_g - \bar{R})/s_R$, normalised by the group mean and standard deviation.

Proposition 1. *If $K_g \equiv K$ within a group, then $\hat{A}_g$ is identical for all $\lambda \geq 0$. The same holds for channel-wise normalisation $\hat{A}_g = z(O)_g - \lambda z(K)_g$ [14] with $z(K)_g := 0$ when $s_K = 0$.*

Proof. $R_g - \bar{R} = O_g - \bar{O}$ and $s_R = s_O$; channel-wise, $s_K = 0$ gives $z(K) \equiv 0$. $\square$

The constraint therefore acts only through groups whose members disagree about the outage count; we call their share the *span rate*. Leakage, being continuous, almost always varies. Under a full-intensity flood ($\rho_t \approx 0.56$), a window becomes a severe outage once $\text{CD}_t > 1 - 0.5/\rho_t \approx 0.10$. This threshold moves with the intensity, which is hard to observe once the link saturates. As GRPO sharpens its policy, whole groups fall beyond the threshold, and only the leakage term keeps steering the update.

Fig. 2 tests this explanation on 80 trained checkpoints under overload. The span rate grows with the within-group spread of the drop action ($r = 0.91$), and checkpoints with a lower span rate suffer more severe outages ($r = -0.61$). Plain GRPO keeps a drop spread of 0.05, spans 47% of groups and causes severe outages in 76% of windows. PPO and Sentinel keep a spread above 0.21 and span every group.

IV. IMBANG

IMBANG reserves one *safe-side member* per group. At each window it takes the policy's sample (τ, f, d) , draws $u \sim \mathcal{U}(0.2, 0.8)$ once per rollout, and executes $\tilde{d} = u d$, $\tilde{\tau} = \tau + (1 - u)(1 - \tau)$ and $\tilde{f} = f$.

Proposition 2. *For identical observations and traffic, the safe-side member's collateral damage after shielding never exceeds*

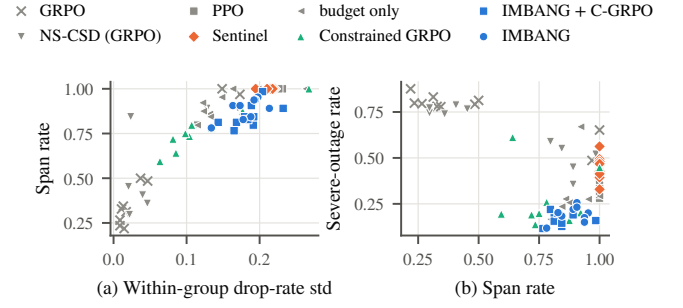


Fig. 2. Constraint cancellation under full-intensity floods (80 checkpoints, $G = 8$ members sharing traffic). (a) The span rate increases with the within-group spread of the drop action. (b) A lower span rate coincides with more severe outages.

Algorithm 1 One IMBANG training iteration

- 1: Sample B levels by MaxMC priority; warm up; fork each into G members sharing one traffic realisation
- 2: Assign policy samples, archive elite, shield probes and the safe-side member
- 3: Roll out H windows through $\mathcal{S}_\phi$; form $R_g = O_g - \lambda K_g$ and group advantages $\hat{A}_g$
- 4: Update θ with the clipped surrogate on policy members, the $[\hat{A}]^+$ -weighted likelihood on the safe member and the elite, and a KL term to the current champion
- 5: Every K iterations: validate, update λ , select the champion feasibility first, and refresh the MaxMC regrets

that of the policy member, so it never incurs more severe outages.

Proof. By (1), CD is non-decreasing in d and non-increasing in τ ; η is unchanged because f is. Each shield rule is a per-coordinate min or max triggered by $(\mathbf{o}_t, f)$ only, so it preserves $\tilde{d} \leq d$ and $\tilde{\tau} \geq \tau$. As ρ_t does not depend on the action, SQ keeps the same order. $\square$

Hence, whenever the sampled actions cause outages that a gentler filter would avoid, the safe-side member supplies the cost variance that Prop. 1 shows to be necessary. The policy learns from it only through $-c[\hat{A}]^+ \log \pi_\theta(\tilde{a} | \mathbf{o})$, that is, only when the member outperformed its group. The guarantee holds at every intensity, so IMBANG never needs to locate the threshold. Three further components complete the method:

- λ follows dual ascent on the validation outage rate, $\lambda \leftarrow [\lambda + \eta_\lambda(\hat{\sigma} - \kappa)]^+$.
- The deployed checkpoint is selected feasibility first.
- Training contexts come from 360 discrete attack levels, prioritised by the MaxMC regret estimate [18], [19] rather than by a learned attacker.

All GRPO variants share the remaining machinery: archive elites, shield probes, a KL anchor to the best validated checkpoint, and Sentinel's two-layer network trained without a critic. Algorithm 1 summarises one training iteration.

Implementation. Each run performs 500 iterations with $B = 32$ contexts, $G = 8$ members, horizon $H = 8$, up to 40 warm-up windows and a discovery step every $K = 20$ iterations. Optimisation uses Adam with learning rate 3×10^{-4} , clipping 0.2, four epochs, a KL weight of 0.02, an elite weight

TABLE II
PROTOCOL B OVER 10 SEEDS: COMPOSITE SCORE ($\uparrow$) AND SEVERE
OUTAGES PER 500 WINDOWS ($\downarrow$)

Scenario	Composite score			Severe outages		
	Sentinel	C-GRPO	IMBANG	Sentinel	C-GRPO	IMBANG
Mild	0.009	0.053	0.040 [†]	0.0	0.0	0.0
Strong	-0.746	-0.612	-0.583 [†]	155.2	85.9	38.9
Chaos/flash	-0.169	-0.114	-0.123 [†]	14.1	16.8	10.7
ICMP (zero-shot)	-0.474	-0.474	-0.458 [†]	4.6	23.3	18.0
Pulsing*	-0.303	-0.228	-0.225 [†]	60.5	37.7	16.1
Polymorphic*	-0.454	-0.442	-0.430 [†]	14.9	24.0	14.2
Boundary*	0.003	0.084	0.071 [†]	0.0	0.0	0.0
ICMP-chaos*	-0.299	-0.255	-0.257 [†]	25.4	25.2	20.9
Average	-0.304	-0.248	-0.246	34.4	26.6	14.8

*Held-out stress test. [†]IMBANG above Sentinel, paired t -test, Holm-corrected, $p < 0.05$. Bold: best.

of 0.5 and an entropy weight of 0.005. The multiplier step is $\eta_\lambda = 4$ per validation round, and the validation battery contains benign, mild, strong and chaos traffic but no ICMP, so that ICMP remains unseen until testing.

V. EVALUATION

We re-implemented Sentinel's public simulator and shield [20] in batched form. Unit tests show agreement with the reference code to within 10^{-5} , and Sentinel's released checkpoints reproduce its published strong-attack results (0.519/0.815/78.2 against 0.519/0.815/75.0 for quality, leakage and severe outages).

Protocols. We compare Sentinel's ten released checkpoints (seeds 42–51) with defenders trained on the same seeds. Protocol B covers eight attack scenarios, four of them held-out stress tests, with 20 episodes of 100 windows each. Protocol A replays the attacker that Sentinel trained for each seed, which is a transfer test for IMBANG. Protocol C is a cross-entropy red team that searches the whole scenario space, ICMP included. All comparisons are paired by seed and traffic. We use paired t -tests with Holm correction and Wilcoxon tests.

Against Sentinel. IMBANG scores significantly higher than Sentinel in every attack scenario (Table II). The scenario average is -0.246 against -0.304 ($p = 1.2 \times 10^{-5}$). Severe outages fall from 34.4 to 14.8 per 500 windows ($p = 0.002$) and leakage from 0.876 to 0.839 ($p < 10^{-3}$), with service quality unchanged. Against Sentinel's own attackers (Protocol A), IMBANG also scores higher (-0.265 against -0.332 , $p = 0.019$) and lets less attack traffic through (0.778 against 0.830, $p = 0.018$); its severe outages drop from 25.1 to 6.3 per 500 windows, although this difference is not significant over ten seeds. In five of the eight Protocol B scenarios (strong, chaos, pulsing, polymorphic and ICMP-chaos), IMBANG is at least as good as Sentinel on quality, leakage and outages simultaneously. The remaining three are trade-offs rather than losses: on mild and boundary-rate floods IMBANG gives up about 0.01 of quality for 4–7 points less leakage. Its red-team worst case improves from -1.07 to -0.72 ($p = 7.2 \times 10^{-4}$), while its variance across seeds shrinks from ± 0.17 to ± 0.03 .

Budget and remedies. A PPO baseline given the same 1.34M training windows already improves on Sentinel, which

TABLE III
ABLATION AND ALTERNATIVE REMEDIES (10 SEEDS). WORST:
RED-TEAM WORST CASE; BUDGET: SEEDS MEETING $\sigma \leq 0.05$

Method	Score $\uparrow$	Severe $\downarrow$	Worst $\uparrow$	Budget
Sentinel [7]	-0.304	34.4	-1.07	–
PPO + shield (equal budget)	-0.276	19.6	-0.79	–
GRPO, no budget	-0.250	91.5	-1.07	–
GRPO + budget	-0.253	36.5	-0.87	1/10
+ entropy bonus $\times 6$	-0.249	33.8	-0.85	0/10
+ uncentred cost	-0.251	34.7	-0.87	1/10
C-GRPO [14]	-0.248	26.6	-0.80	3/10
IMBANG	-0.246	14.8	-0.72	9/10
IMBANG + C-GRPO normalisation	-0.247	12.1	-0.71	7/10

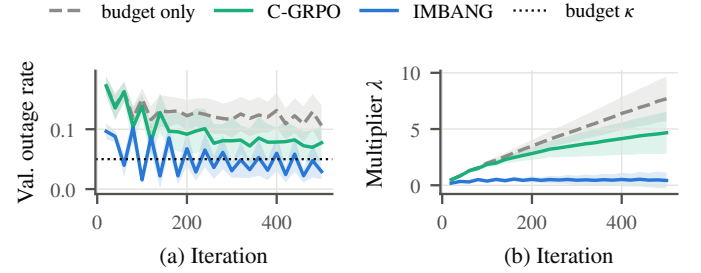


Fig. 3. Training dynamics (mean and 95% confidence band over ten seeds). (a) Validation outage rate. (b) Lagrange multiplier. Without a safe-side member, raising λ cannot pull the outage rate below the budget; IMBANG meets it with $\lambda \approx 0.4$.

shows the benefit of dropping co-evolution. IMBANG still beats it at equal budget ($+0.030$, $p = 2 \times 10^{-7}$). Table III and Fig. 4 isolate the safe-side member:

- A Lagrangian budget alone is met in one seed out of ten even though λ rises to about 7.7, as Prop. 1 predicts (Fig. 3).
- An entropy bonus or an uncentred cost does not restore feasibility.
- Constrained GRPO meets the budget in three seeds.
- IMBANG meets it in nine with $\lambda \approx 0.4$, matching Constrained GRPO's score ($p = 0.47$) with fewer outages (14.8 against 26.6) and slightly more leakage (0.839 against 0.820).

Deployment and limitations. The deployed actor has 5,447 parameters (21.8 kB). On one 2.1 GHz core, a decision including shielding takes 0.24 ms, far below sub-second control windows. The study has four limitations:

- On zero-shot ICMP floods, IMBANG incurs more severe outages than Sentinel (18.0 against 4.6, not significant). A simulated operator who reviews the link every ten windows and applies Sentinel's ICMP constants removes this gap: 2.1 severe windows at 90% recognition accuracy and 4.3 at 70%, with no change in the composite score.
- IMBANG does not surpass Constrained GRPO on the composite score; its advantage lies in meeting the outage budget reliably and in fewer outages.
- Prop. 2 assumes that congestion is evaluated on the offered load.
- All results come from simulation.

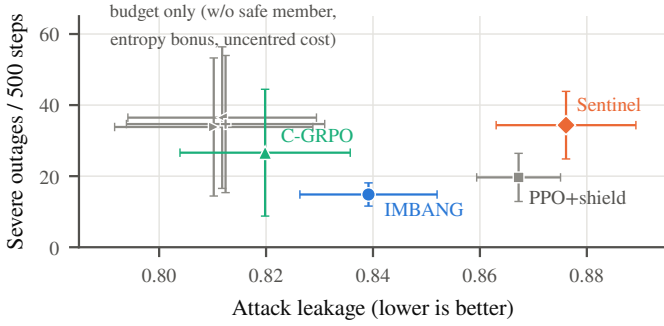


Fig. 4. Severe outages against attack leakage (Protocol B, eight scenarios, 95% confidence intervals over ten seeds). Only IMBANG reduces both relative to Sentinel.

VI. DISCUSSION

Why a single member suffices. Fig. 3 shows that the multiplier is not the bottleneck. Without a safe-side member, λ grows steadily, yet the outage rate barely moves, because most groups contain no member that avoids the outage. One member that is guaranteed to be no less safe is enough to give the constraint a direction, so IMBANG meets the budget from the first hundred iterations while λ stays almost constant. Derived from the policy’s own sample, it needs no external teacher and no hyperparameter beyond the range of u , and it replaces one of the eight members rather than adding a rollout, so a training iteration costs no more than before.

Interpretability. IMBANG keeps Sentinel’s operator-facing rule extraction. Across the ten seeds, the depth-4 decision trees distilled from the final policies reproduce its filtering focus with 99.8% held-out accuracy, and its threshold and drop rate with mean absolute errors of 0.013 and 0.017. They use about 32 leaves in total. Deployed behind the same shield, these rule programs reach the same validation score as the neural policy (0.089), so an operator can audit or replace the network without loss. The certified shield tuning is equally readable. On average it lowered the PanicGuard utilisation gate from 0.60 to 0.52, while keeping the drop cap and the threshold floor close to Sentinel’s values (0.11 and 0.39 against 0.10 and 0.40). Low-rate floods can therefore be filtered slightly earlier without any change to behaviour on benign traffic.

Scope of the guarantee. Prop. 2 needs two structural facts: collateral damage must be monotone in the drop rate and the threshold, and the shield must act through coordinate-wise clamps. Both hold for rate-limiting filters in general, but the second part of the proof also uses the fact that congestion depends on the offered load. If filtering instead took place upstream of the bottleneck, weaker filtering could itself overload the link, and the safe direction would have to be derived from the joint effect on congestion. Extending the construction to that case, and to trace-driven 5G and 6G traffic, is our next step.

VII. CONCLUSION

Critic-free group-relative training removes the value network and the adversarial loop from DDoS defence, but it erases any safety cost that a whole group shares. A single

safe-side member per group, guaranteed by monotonicity, restores that cost. The resulting defender outperforms the co-evolutionary Sentinel and more than halves its severe outages. Validation on replayed real traffic, such as captures from live 5G testbeds, is left for future work.

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